Video title: Building Image Datasets in PyTorch (Module 2)

Code snippets reference

This document lists the code snippets shown in the video along with brief descriptions. We recommend reviewing this document alongside the video for the best learning experience.

Disclaimer:
The code snippets shown in this video are for conceptual illustration only. They are simplified or partial examples designed to explain the underlying ideas and may not run as-is. Please avoid copying the code directly without understanding the context and structure required to make it work.

<table>
<tr><th>S. No.</th><th>Code</th><th>Description</th></tr>
<tr><td>1.</td><td><pre>import torchvision
from torchvision import datasets, transforms

dataset = datasets.FashionMNIST(
    root="data",
    train=True,
    download=True,
    transform=transforms.ToTensor()
)</pre></td><td>Slide title: Loading a built-in dataset
Purpose: Loads the Fashion-MNIST dataset with automatic download and converts images to tensors for model training.</td></tr>
<tr><td>2.</td><td><pre>mnist = datasets.MNIST(
    root="data",
    train=True,
    download=True,
    transform=transforms.ToTensor()
)</pre></td><td>Slide title: Loading a built-in dataset
Purpose: Loads the MNIST handwritten digit dataset using Torchvision for quick experimentation.</td></tr>
<tr><td>3.</td><td><pre>image, label = dataset[0]
print(label)</pre></td><td>Slide title: Accessing image samples
Purpose: Retrieves the first image and label from the dataset and prints the label.</td></tr>
<tr><td>4.</td><td><pre>from torch.utils.data import Dataset

class ImageDataset(Dataset):

    def __init__(self, dataset, transform=None):
        self.dataset = dataset</pre></td><td>Slide title: Creating a custom image dataset class
Purpose: Defines a custom dataset class that</td></tr>
</table>

	`` self.transform = transform ```	stores an image dataset and optional preprocessing transforms.
5.	``` def __getitem__(self, index): image, label = self.dataset[index] if self.transform: image = self.transform(image) return image, label ```	• **Slide title:** Creating a custom image dataset class • **Purpose:** Specifies how each dataset sample is retrieved and optionally transformed before being returned.
6.	``` transform = transforms.Compose([transforms.CenterCrop(20), transforms.ToTensor()]) ```	• **Slide title:** Applying Torchvision transforms • **Purpose:** Creates a preprocessing pipeline that crops images and converts them into tensors using Torchvision transforms.